Week 11 — Turtle Geometry

Name: ____ Date: ____

> *The turtle turns the **outside** of every corner — and one full lap around any polygon is always **360°** of turning. Turn at each corner = **360 / n**.*

Today's words

Word	What it means
turtle	our drawing robot — it draws everywhere it walks
toolbox	nine little function files that ARE the turtle — you can read them!
`forward(n)` / `right(d)`	walk n steps / spin d degrees (turning never draws)
`penup` / `pendown`	lift the pen to move invisibly / put it back down
exterior angle	the turn at a corner — the outside; what the turtle turns
interior angle	the angle inside the corner — always 180 − the turn

1 · Be the turtle 🐢

Your pencil is the pen. Start at **S**, facing **→**. One grid square = 20 steps. Draw the turtle's path.

```
forward(60)
right(90)
forward(40)
right(90)
forward(60)
```

```
S .  .  .  .  .

.  .  .  .  .  .

.  .  .  .  .  .

.  .  .  .  .  .
```

What familiar shape is it — and which side is missing? ____

2 · The polygon table 📐

Fill in the missing values. (Turn = 360 / n. Interior = 180 − turn.)

sides n	turtle's turn (exterior)	interior angle
3	______	60°
4	90°	90°
5	72°	______
6	______	______
8	______	______

3 · Spot the bug 🐞

This was supposed to draw a square. Circle the bug, then answer below.

```
turtle_start
for side = 1:4
    forward(100)
end
right(90)
```

What does it draw instead? ____

🧠 Brain teaser (optional — take it home)

Write the turtle commands to draw a **rectangle 200 steps wide and 100 steps tall** — *or* the **first letter of your first name**. Rules: only `forward( )`, `backward( )`, `right( )`, `left( )`, `penup`, and `pendown`, one command per line. Trace it with your finger to test — you are the turtle. Bring it next week for a shout-out (we may run it on the smartboard!).